

3D documentation certification methodology

aPrehliadky · methodology version 4 (31 October 2026, aPrehliadky 3.2.5 / Fotograme3ka 3D 2.1.3) · forensika.eu

The 3D documentation certificate records how the scan was made (capture trace), what substantiates its scale (verifications) and exactly what it certifies (file hashes). The app issues it for a finished model; it exists as certificate.json in the .f3d package and as a PDF for the case file. The certificate computes nothing — every value in it comes from the session manifest or from a file hash.

The certificate is not a metrological calibration of the device and does not replace expert judgement. It states by how many independent routes the model scale is substantiated, and with what result.

1. Sources of scale and their independence

Source	What it measures	Independent of
LiDAR depth (time-of-flight)	distance of every image point from the camera; photogrammetry receives a depth prior — it does not guarantee the model scale (audit 3.0: the same scan came out 6 % and 14 % smaller)	photogrammetry, the operator
Camera trajectory fit (ARKit → model)	similarity transform between camera positions from ARKit (IMU + visual odometry + LiDAR, metric) and poses from photogrammetry; coefficient 1/k, uncertainty u(k) = √((σ/(r·√n))² + max(2 %, drift)²), drift = standard deviation of the fit scale over 3–6 time windows (v4; up to v3 the floor was 0.5 %, real scans were 2–5 % off: test12 window drift 1.08 → 1.02)	LiDAR depth in photogrammetry, the target
LiDAR depth vs. model (v4)	frame depth (time-of-flight) vs. the camera → model surface distance along the same ray; robust line model = k · LiDAR + b through thousands of points from hundreds of frames (0.3–1.5 m band, no edges, rejection |r| > 3·MAD) plus the slope without offset; u(k) = √(jackknife² + 2 %² + (½ slope difference)²); conclusive with ≥ 20 frames, ≥ 500 points, u < 3 %, |b| < 60 mm. Free of the trajectory drift (test12: k ≈ 1.00 vs. trajectory 1.054) but shares the LiDAR calibration (test5 in sunlight: depth and trajectory both 4.8 % off the target)	ARKit drift, the target, the operator; not of the LiDAR calibration
T3 target in the model	the centres of circular fiducials (Ø 24 mm, base 247 × 160 mm) are found in the frames and triangulated from the photogrammetry poses directly in the model frame; 10 distances between circles A–E (33–294 mm) vs. the reference × print scale → k. From 3.2: L² weights, rejection of a fiducial whose pairs are systematically off (the QR centre “3Q” ~10 mm off on a real scan), uncertainty from the number of independent points; conclusive only with a ratio scatter < 3 %	LiDAR and the ARKit trajectory; verifies photogrammetry
Reference	the operator measures a segment of known length in the model (target mm scale 200/100 mm, centres A–C 294 mm, ruler, tape) and enters the true value; verdict with a tolerance of max(1 %, U of the reference measurement); with several references the one with the largest deviation is decisive	LiDAR, ARKit; depends on the operator
Target (LiDAR) — T2 and T3	the fiducial centres are computed in 3D from LiDAR depth and the camera pose; the distances are compared with the printed reference	photogrammetry, the operator; verifies the LiDAR + ARKit chain

The measurement scale is the fusion of the trajectory, LiDAR depth vs. model (v4) and the target in the model (inverse-variance weighting) with a consistency check (from 3.2): Birge ratio $\sqrt{(\chi^2/(n-1))} \le 2 \rightarrow$ weighted mean; $> 2 \rightarrow$ the sources are statistically incompatible (test5: trajectory 1.0270 vs. target 0.9788 ± 0.9 %, Birge > 2 — the mean would be off the physical target) and the source with the best traceability decides: the target in the model, without a target the LiDAR depth; the trajectory remains a check, the scale uncertainty = max(u_decisive, d/2). The trajectory and the LiDAR depth share the LiDAR metre, so against the target they act as one source (weighted mean with the uncertainty of the better one) and cannot outvote the target. Every measurement carries an expanded uncertainty U (k = 2) from the scale uncertainty, the point placement (σ of the pointing method × mm per screen point at placement; floor finger 1.7 mm, Apple Pencil 0.5 mm, mouse 0.2 mm), the frame resolution and, from 3.2, the model resolution (texture texel, mesh facet). A volume has an uncertainty only for a closed mesh; an open mesh gives an indicative value. The T3 target also serves validation of the app and the device: same target, same procedure, many scans and devices → the error distribution is material for the validation study (docs/VALIDACIA_protokol.md). The print is verified with a caliper (centres A–B = 247.00 mm) and the value is entered in the app.

2. Documentation class

Class	Conditions	Meaning
A — metrologically verified	LiDAR depth for ≥ 90 % of frames and scale corrected by a metrology-grade trajectory fit (residual < 25 mm, ≥ 50 cameras) and an independent verification able to resolve 1 %: a reference ≥ 290 mm (points placed by Fine-tune or the Pencil) or a conclusive target in the model with consistent sources — the worst of the available verification errors ≤ 1 % (or two independent verifications ≤ 1 %); no verification above 2 %	measurements substantiated by two independent routes; values binding within the uncertainty U
B — metric	scale corrected by a metrology-grade trajectory fit, a conclusive LiDAR depth vs. model (v4) or a conclusive target in the model, or LiDAR scale confirmed by a reference/target; independent verification missing, shorter than 290 mm, or ≤ 2 %	the model is metric, a second proof is missing — state “no reference” in the report
C — indicative scale	scale known only approximately: LiDAR without a fit correction (really a 6–14 % error) or a coefficient from the trajectory without LiDAR; verification ≤ 5 %; or a reference that disagrees ($\lvert\Delta\rvert$ > max(1 %, U))	values indicative; add or repeat a reference before citing in an expert opinion
D — unverified scale	scale not substantiated (no LiDAR and no trustworthy fit) or verification > 5 %	illustrative documentation only

“Trustworthy fit” = residual / scale < 150 mm, ≥ 8 cameras (enough for cropping). “Metrology-grade fit” = residual / scale < 25 mm, ≥ 50 cameras (enough to correct the scale; real scans: 7.6–14 mm with 158–493 cameras; test12 with a 27.5 mm residual had a trajectory whose scale floated by ±3 %, so the absolute threshold stays). LiDAR depth vs. model is not an independent verification for A (LiDAR is not a physical reference, floor 2 %); it gives a metric scale, class B. “Independent verification” = the worst of the reference error, the LiDAR target error and the difference between the target in the model and the scale in use (the best of them would mask a failure of the others); the LiDAR target counts with ≥ 3 distances and ≥ 5 observations per fiducial. A reference shorter than 290 mm cannot resolve 1 % (with a finger U ≈ 2.7 % on 200 mm) and confirms at most B. After a recompute (iPhone or Mac) the references and the certificate of the previous model are set aside (previousScaleChecks, previousCertificates) and the new model starts unverified — up to 3.1.4 the class was computed from the old reference. The class is computed by the same function in the report and in the certificate, so the two documents are always consistent.

3. Capture trace

From the frame metadata (camera pose, time, depth) the certificate states: start and end, duration, number of frames and frames with depth, resolution, coverage (azimuths of 36, elevation bands of 3), camera path length, range and mean of the camera distance from the scan centre, number of frames with a validly found target, sharpness and rejected blurred frames. From 3.2 the report also states session interruptions (a call, another app) and whether the position was recovered by relocalisation — frames after an unrecovered interruption may be in a different coordinate frame. The PDF shows a top view: camera path, frames, scan centre and the 3D fiducial positions. The trace shows that the scan was captured according to the procedure (three heights, full circle, appropriate distance) — and exposes one that was not.

4. Certified content and integrity

- SHA-256 of the model (model/model.usdz), the room mesh (model/room.usda) and the findings (annotations.json) at the time of issue; number of findings and measurements. From 3.2 the report also carries the SHA-256 of the manifest.
- SHA-256 of the archive package, if the scan was archived.
- Reconstruction record (3.2): app and version, platform (iOS/macOS), device, memory, detail, input, frames used and dropped, input hash verification on the Mac, stages and duration — the report distinguishes “Captured with” and “Computed with”.
- payloadSHA256 = SHA-256 of the canonical certificate bytes (sorted keys, ISO 8601 dates, no whitespace; slash as \/). Any change to the certificate invalidates the hash. The certificate structure does not change in 3.2.
- Signature (Pro; from Fotograme3ka 3D 2.1 also with the Mac’s key): ECDSA P-256 / SHA-256 with the device key (Secure Enclave where available) over the canonical bytes; the certificate carries the public key (X9.63), the signature (DER), time, signer, institution and the key fingerprint (first 8 bytes of the SHA-256 of the public key).
- Identifier F3D-XXXXX-XXXXX = SHA-256(session ID + model hash + issue time). QR code in the PDF: F3D:<ID>:<class>:<model hash prefix>:<session ID>.

5. Verifying a certificate

1. In the app: import the .f3d package. The import verifies the hashes of all files (from 3.2 also missing and unlisted files), the package signature (if any) and the certificate: content integrity, the match of the model hash with the model in the package and the certificate signature. The result is in the session overview (“Verified · signed” / “Hashes match — origin not substantiated” / “CHANGED”) and in the report.
2. Without the app: unzip -p scan.f3d model/model.usdz | shasum -a 256 must give the hash from the certificate; unzip -p scan.f3d certificate.json contains the certificate, its payloadSHA256 and the signature; any ECDSA P-256 library verifies the public key and signature over the canonical bytes.
3. Compare the QR code in the PDF with the ID and hash in the package.

6. Device validation — recommended procedure for a study

1. Print the T3 target at 100 % (A4 landscape), mount it on a rigid board, check the circle centres A–B (247.00 mm) with a caliper and enter the value in the app; a print deviation > 0.5 mm → new print.
2. Target flat, matte background, diffuse light. Capture ≥ 10 scans with the Object profile, extent Detail, distance 0.3–0.6 m, three heights, on each device.

3. From each scan export the Validation CSV (Export → Validation CSV): scale and uncertainty from the trajectory, from the target in the model, the Birge ratio and difference, rejected fiducials, the LiDAR target error, sharpness, class and all measurements with U.
4. Evaluate mean, SD and the 95th percentile of errors by device and distance; the class A criterion (≤ 1 %) should be met in ≥ 95 % of scans.
5. Validate larger dimensions (body length) with a 500–1000 mm reference (measuring tape, two marks) — reference in the report.

7. Limitations

- LiDAR depth has a resolution of 256 × 192 px; the fiducial centre accuracy is ~1–3 mm at 0.5 m and decreases with distance and oblique views. Hence the median over many frames and outlier removal.
- The LiDAR target verifies the LiDAR + ARKit chain; the scale of the photogrammetric model comes from the trajectory fit, LiDAR depth vs. model and the target triangulated in the model (fusion with a consistency check). The certificate reports all routes separately.
- The ARKit trajectory can change its scale by 2–5 % during a capture (fast motion, blurred frames, close to a small object) and outdoors in sunlight the LiDAR measured 4.8 % differently from the physical target (test5). Scans without a target receive a metric scale from the LiDAR depth (B) but no physical verification; for the validation study and class A the T3 target in view is essential.
- The centre of the target's QR code (“3Q”) is not a circular fiducial — its triangulation is worse and from 3.2 it is rejected as an outlier; the scale comes from circles A–E.
- The colour patches of the target are indicative; colour-binding documentation requires a commercial ColorChecker.
- The certificate confirms the state at the time of issue; later findings and measurements change the findings hash — issue a new certificate after edits. After a recompute a new reference and a new certificate are needed.